www.Padasalai.Net www.TrbTnpsc.com

Tsi12Bio

Tenkasi District
Common Quarterly Examination - September 2025

Standard 12

Time Allowed: 3.00 Hours **BIOLOGY** Maximum Marks: 70

BIO-BOTANY

Marks: 35

I. Choose the best answer: **8×1=8**

1) An eminent Indian Embryologist is _____.
   a) S.R. Kashap b) P. Maheswari
   c) M.S. Swaminathan d) K.C. Mehta
2) In Haplopappus gracilis, number of chromosome in cells of nucellus is 4. Will be the chromosome number in primary endosperm cell?
   a) 8 b) 12 c) 6 d) 2
3) Which of the following is an example of polygenic inheritence?
   a) flower colour of Mirabilis jalapa b) production of male honey bees
   c) pod shape in garden pea d) skin colour in humans
4) In his classic experiments on pea plants, Mendel did not use _____
   a) flowering position b) seed colour
   c) pod length d) seed shape
5) Due to incomplete linkage in maize, the ratio of parental and recombination are _____
   a) 50:50 b) 7:1:1:7 c) 96:4:3:6 d) 1:7:7:1
6) If haploid number in a cell is 18. The double monosomic and trisomic number will be _____
   a) 34 and 37 b) 34 and 35 c) 37 and 35 d) 17 and 19
7) pBR 322, BR stands for _____
   a) plasmid bacterial recombination b) plasmid bacterial replication
   c) plasmid boliver and rodriguez d) plasmid baltimore and rodriguez
8) Soliditying agent used in plant tissue culture is _____
   a) Nicotinic acid b) Cobaltous chloride c) EDTA d) Agar

II. Answer any 4 of the following questions: **4×2=8**

9) What do you mean Embryoids? Write its application.
10) Define the term diplospory.
11) What is meant by hybrid vigour?
12) What is gene mapping? Write it uses.
13) What is meant by synapsis?
14) What are the materials used to grow micro organism like spirulina?

III. Answer any three questions and question number 19 is compulsory: **3×3=9**

15) Note on Taq polymerase and bioleaching.
16) Differences between missense and nonsense mutation.
17) Name the seven contrasting characters of Mendel.
18) List out the functions of tapetum.
19) Write a note on knop's solution and its composition.

IV. Answer all the questions: **2×5=10**

20) a) Give a concise steps involved in fertilization of an angiosperm plant.

(OR)

b) Explain the Incomplete dominance.

21) a) Draw the diagram of different types of Aneuploidy.

(OR)

b) Write down the application of single cell protein.

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

BIO - ZOOLOGY **Marks: 35**

I. Choose the best answer: **8×1=8**

1) Increased Androgen , excessive facial (or) body growth, acne is the symptom of _____.
   a) Candidiasis b) Genital Warts c) AIDS d) PCOS
2) Parthenogenetic egg may develop into individuals of any sex called _____
   a) Amphitoky b) Thelytoky c) Arrhenotoky d) none
3) _____ are commonly called as international disease.
   a) Hepatitis and Syphilis b) Genital Wart and AIDS
   c) Candidiasis d) Syphilis and Gonorrhoea
4) XX-XO type of sex determination observed in
   a) bugs, cockroach and grasshoppers b) honey bees
   c) drosophila d) gypsy moth and birds
5) The first codon to be deciphered was _____ which codes for _____.
   a) AAA, Proline b) GGG, Alanine
   c) UUU, Phenylalanine d) TTT, Arginine
6) Age of fishes was _____.
   a) Silurian b) Permian c) Devonian d) Cambrin
7) Thick Mucopurulant nasal discharge is the symptoms of _____.
   a) Pneumonia b) Common cold c) Tuberculosis d) Diphtheria
8) Alcohol content in wine is _____
   a) 3-5% b) 9-14% c) 4-6% d) 35-50%

II. Answer any four questions: **4×2=8**

9) What is bioremediation ?
10) What is meant by placentation?
11) Symptoms of genital herpes.
12) What are holandric genes?
13) Why tRNA is called an adapter molecules?
14) Note on atavistic organs.

III. Answer any three questions. Question number 19 is compulsory: **3×3=9**

15) Name the viral diseases are classified on the basis of the symptoms produced in the body organs.
16) Write a note on cyclosporin - A.
17) What is male heterogamety?
18) Draw and label the human ovum.
19) Differentiate divergent evolution and convergent evolution.

2×5=10

IV. Answer all:

20) A brief account about menstrual cycle.

(OR)

What is meant by Karyotyping? How it can be prepared? Mention its uses.

21) Explain the human evolution. **(OR)**
How the drugs are classified? Mention its effects.

C. Christo
M.A B.Ed
Tenkasi

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com